

4.2 The Mean Value Theorem

The Mean Value Theorem is one of the very important theorems in Calculus, for two reasons:

1. This theorem is used widely in Mathematics to prove other theorems and
2. The theorem is a useful tool to solve many practical problems

In Section 4.1 we dealt with two theorems: The Extreme Value Theorem and Fermat's theorem.

Apart from the fact that these two theorems helped us to devise a method to find the extreme values of a function, these theorems also help us to prove more theorems. One of these is another important theorem that we will discuss in this section, namely Rolle's Theorem, which you can find on page 287 of the text book.

Rolle's Theorem

Let f be a function that satisfies the following three conditions:

1. f is continuous on the closed interval $[a, b]$
2. f is differentiable on the open interval (a, b)
3. $f(a) = f(b)$

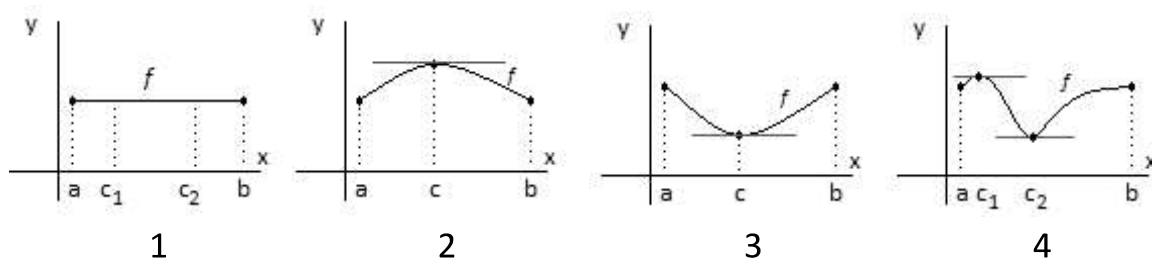
Then there is a number c in (a, b) such that $f'(c) = 0$.

What is the significance of $f'(c) = 0$?

$f'(c) = 0$ if and only if the slope of the tangent to the curve is 0 at the point where $x = c$, that is,

$f'(c) = 0$ if and only if the tangent to the curve is parallel to the x -axis at the point where $x = c$.

Here are some graphs of functions that show different scenarios where Rolle's Theorem can be applied:



Note that in each case, the function f is continuous on the interval $[a, b]$, f is differentiable on the interval (a, b) and $f(a) = f(b)$.

The function in sketch 1 is a constant function: In this case $f'(x) = 0$ for all $x \in (a, b)$ and so c can be any number in the interval (a, b) , as is indicated by the numbers c_1 and c_2 in the sketch. Note that in this case the tangent to the graph of f is parallel to the x -axis at every point in (a, b) .

As can be seen from sketch 4, even for a function that is NOT a constant function, there can be more than one number in (a, b) where the tangent line is parallel to the x -axis. Rolle's theorem guarantees us that if a function f satisfies the three conditions of the theorem, then there will be at least one number $c \in (a, b)$ such that $f'(c) = 0$.

Example 1: Consider the function f defined by $f(x) = x^3 - 2x^2 + x$, $x \in [0, 1]$

We Apply Rolle's theorem to find the points where the tangent to the graph of f is parallel to the x -axis.

Solution : Since f is a polynomial function, we can conclude that

1. f is continuous on $[0, 1]$
2. f is differentiable on $(0, 1)$
3. $f(0) = f(1) = 0$ (Check the calculations!)

So, according to Rolle's theorem there exists a number $c \in (0, 1)$ such that $f'(c) = 0$. Let's see if we can find this number c :

Note that $f'(x) = 3x^2 - 4x + 1$. So, $f'(x) = 0 \Rightarrow 3x^2 - 4x + 1 = (3x - 1)(x - 1) = 0$

Thus, $f'(x) = 0 \Rightarrow x = 1/3$ or $x = 1$.

Thus, the number $c = 1/3 \in (0, 1)$ is such that $f'(c) = 0$, that is, at $x = 1/3$ the tangent to the graph of f is parallel to the x-axis.

Here is another problem that you can solve by using Rolle's Theorem: The solution will be discussed during the online lecture.

Example 2: Consider the function f defined by $f(x) = e^x + x^3$, $x \in \mathbb{R}$.

1.1 The Intermediate Value Theorem states that if a function f is continuous on a closed interval

$[a, b]$, with $f(a) \neq f(b)$ and q is any number between $f(a)$ and $f(b)$, then there exists a number c in (a, b) such that $f(c) = q$.

Use this theorem to show that $f(x) = 0$ has at least one root.

Hint: Consider f on the interval $[-2, 2]$.

1.2 Use Rolle's Theorem to show that $f(x) = 0$ has exactly one root.

The Mean Value Theorem (Page 288 of the text book)

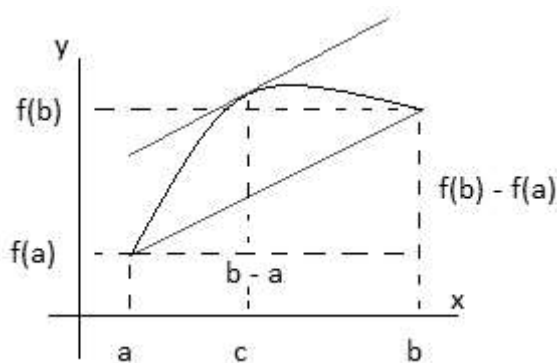
Let f be a function that satisfies the following two conditions:

1. f is continuous on the closed interval $[a, b]$
2. f is differentiable on the open interval (a, b)

Then there exists a number $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$

Interpretation of the Mean Value Theorem:

Note that the expression $\frac{f(b) - f(a)}{b - a}$ is the slope of the line segment that joins the two points $(a, f(a))$ and $(b, f(b))$ on the graph of f , as in the following sketch:



Thus, the Mean Value Theorem states that (if the two conditions are met), at some point in (a, b) , the tangent line to the graph of the function will be parallel to the line segment that joins the points $(a, f(a))$ and $(b, f(b))$.

Example 3: Consider the function f defined by $f(x) = x^2 + 5x$, $x \in [-5, 2]$.

Show that f satisfies the conditions of the Mean Value Theorem and find a suitable number " c " guaranteed to exist by the Mean Value Theorem.

Solution:

1. Since f is continuous on $\mathbb{R}$ it follows that f is continuous on $[-5, 2]$
2. Since f is differentiable on $\mathbb{R}$ it follows that f is differentiable on $(-5, 2)$

Thus, by the Mean Value Theorem, there exists a number $c \in (-5, 2)$ such that:

$$f'(c) = \frac{f(2) - f(-5)}{2 - (-5)} = \frac{14 - 0}{7} = 2$$

But $f'(x) = 2x + 5$, so $f'(c) = 2 \Rightarrow 2c + 5 = 2 \Rightarrow c = -3/2$. Note that $-3/2 \in (-5, 2)$.